Unit-4 Network Security

4.1 Security topologies – security zones, DMZ, Internet, Intranet, VLAN, Security implication, tunneling.

❖ **Internet**

- ✓ The internet is a worldwide connection of networks.
- ✓ It is accessible to every user all over the world.
- ✓ The internet is the name given to the entire public network which provides the infrastructure for the transfer of data between remote points.
- ✓ Such data can take the form of email, web pages, files, multi-media and just about anything else that exists in digital form.
- ✓ Internet use the standard Internet Protocol.(TCP/IP)

❖ **Intranet zone:**

- ✓ Multiple PCs are connected with each other.
- ✓ PC in intranet is not available to the world outside of the intranet.
- ✓ It is a trusted area of network.
- ✓ An intranet can be described as a mini1-internet build within the safety of a secure networking environment.
- ✓ Example: Each company or organization has their own Intranet network and employees of that company can access the computers.
- ✓ Intranets are typically used to provide internal corporate web sites for employee only access. Because the intranet servers have internal, private IP addresses and reside behind firewalls they are generally not accessible to the outside world.

❖ **VLAN** -Virtual local area networks

- ✓ Computers and users are grouped logically instead of physically.
- ✓ It involves grouping devices on a single switch into multiple broadcast domain and network segments.
- ✓ This provides a way to limit broadcast traffic on each segment of network (Improving overall performance) and increase security through the deployment of multiple isolated LAN on a single switch.

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- ✓ Trunkling: It is used to create a VLAN which span multiple switches. This enabled user to be grouped on VLAN based on function rather than physical location.
- ✓ Example: all members of accounting department can be placed in same VLAN instead of physical location or physical connection.
- ✓ A way of dividing a single physical network switch among multiple network segments or broadcast domains.
- ✓ Routers are needed to make different VLANs to talk

The purpose of a VLAN: Network engineers use VLANs for multiple reasons, including the following:

- to improve performance
- to tighten security
- Cost Reduction

Improve performance: It reduce unnecessary traffic on the network.

Tighten security: Groups that have sensitive data are separated from the rest of the network.

Cost reduction: More efficient use of existing bandwidth and less need for expensive network upgrades.

Types of VLANs

Port-based VLAN: Use the physical layer port number to assign computers to VLAN segments.

MAC based VLAN: Use the data link layer addresses to form the VLAN.

IP based VLAN: Use the Network layer addresses to form the VLAN.

❖ **DMZ-Demilitarized Zone**

- ✓ In computer networks, a DMZ is a computer host or small network inserted as a "neutral zone" between a company's private network and outside public network.
- ✓ It prevents outside users from getting direct access to a server that has company data.
- ✓ A DMZ is an optional and more secure approach to a firewall and effectively act as a proxy server as well.
- ✓ **Example**: In a DMZ configuration for a small company, a separate computer receives a request from users within the private network for access to websites or other public network.
- ✓ Then DMZ host send these request on the public network.
- ✓ The DMZ may have the company's web pages so that these could be served to the outsider world. The DMZ does not provide access to other company data.

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- ✓ Any services that is being to provided to users on the external network can be placed in the DMZ.
- ✓ In the event that an outside user penetrated the DMZ host's security, the web pages might be corrupted but no other company information would be exposed.
- ✓ Services of a DMZ include:
  - DNS servers
  - FTP servers
  - Mail servers
  - Proxy servers
  - Web servers

❖ **Tunneling:**

- ✓ Tunneling is a protocol that allows for the secure movement of data from one network to another.
- ✓ Tunneling involves allowing private network communications to be sent across a public network, such as the Internet, through a process called encapsulation.
- ✓ In tunneling, the data are broken into smaller pieces called packets as they move along the tunnel for transport.
- ✓ As the packets move through the tunnel, they are encrypted and another process called encapsulation occurs.
- ✓ The private network data and the protocol information that goes with it are encapsulated in public network transmission units for sending.

Example:

- ✓ A company has multiple locations and decides to use the public Internet to connect the networks.
- ✓ To make this connection secure from outside unauthorized use, the company can employ a VPN connection between the different networks.
- ✓ On each network, an edge device usually a router or VPN concentrator connects to another edge device on the other network.
- ✓ Then using IPSec protocols, these routers establish a secure, encrypted path between them.
- ✓ This securely encrypted set of packets cannot read by outside routers.

4.2 Firewalls: Need of Firewall, Working of Firewall, Types of Firewall: Packet Filtering, Stateful Inspection, Application Level Gateway, Circuit-Level Gateway and Next-Generation Firewall

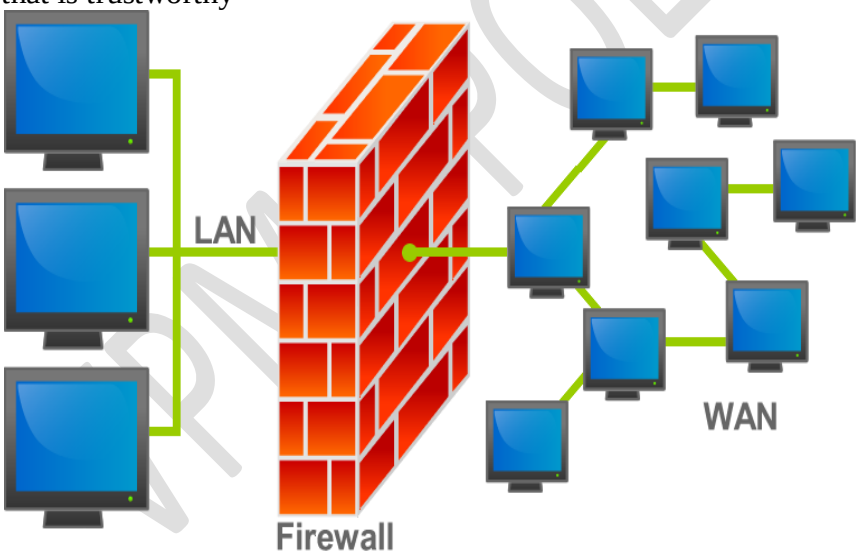
Firewall: A firewall is software or hardware based network security system that controls the incoming and outgoing network traffic based on some rules.

- ✓ In general, firewalls are installed to prevent attacks.

Need of Firewall

- ✓ **Monitoring Network Traffic :**Firewall security starts with effective monitoring of network traffic based on pre-established rules and filters to keep the systems protected. Monitoring of network traffic involves the following security measures.
  - Block incoming network traffic based on source or destination
  - Block out going network traffic based on source or destination
  - Block incoming network traffic based on content.
  - Report on Network traffic & firewall activities

- ✓ **Stops Virus Attacks and spyware:** Firewalls are a crucial line of defense against malicious software.
- ✓ **Preventing Hacks:** Cyber threats are evolving at a fast pace and are widespread. emails, systems, and other sensitive information. A firewall can either entirely block a hacker
- ✓ **Promotes Privacy :** Having a firewall keeps the data safe and builds an environment of privacy that is trustworthy



Types of firewall

1) Packet filters

- ✓ This firewalls work at the network layer of the OSI model or IP layer of the TCP/IP.
- ✓ They are usually a part of router.

- ✓ Here each packet is compared to a set of criteria before it is forwarded.
- ✓ On the Internet, packet filtering is the process of passing or blocking packets at a network interface based on source and destination addresses, ports, or protocols.

Functionality:

- ✓ It receives each packet when it arrives.
- ✓ It apply rules based on content of packet information.
- ✓ If any match found then it will decide either to keep or discard the packet based on rules.
- ✓ If no match found then it will take default action either by discard or accept.

Advantages:

- ✓ It is simple to implement.
- ✓ Fast in their operating speed.
- ✓ It has low cost and low impact on network performance.

Disadvantage:

- ✓ Difficult in setting rules correctly.
- ✓ Need authentication support.

2) Circuit level gateways

- ✓ It works at Session Layer of OSI model or TCP Layer of TCP/IP.
- ✓ They are usually a part of router.
- ✓ Here each packet is compared to a set of criteria before it is forwarded.
- ✓ They Monitors TCP handshaking between packets, to determine a requested session is legal or not.
- ✓ It is relatively inexpensive.
- ✓ They do not filter individual packets.

Advantage:

- ✓ Most secure method.
- ✓ Does not provide end to end connection.

Disadvantage:

- ✓ Time consuming with respect to previous method.

3) Application level gateways

- ✓ Also called Proxies, similar to circuit-level gateway except that they are application specific.

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- ✓ It works at Application Layer of OSI model
- ✓ Here each packet is compared to a set of criteria before it is forwarded.
- ✓ Some of the services supported by application gateway are FTP,SMTP,POP3,HTTP etc.
- ✓ They examine packets at application layer, so they can filter application specific commands such as http:POST and GET etc…
- ✓ It has full access to protocol:
  - user requests service from proxy
  - proxy validates request as legal
  - then actions request and returns result to user

Functionality:

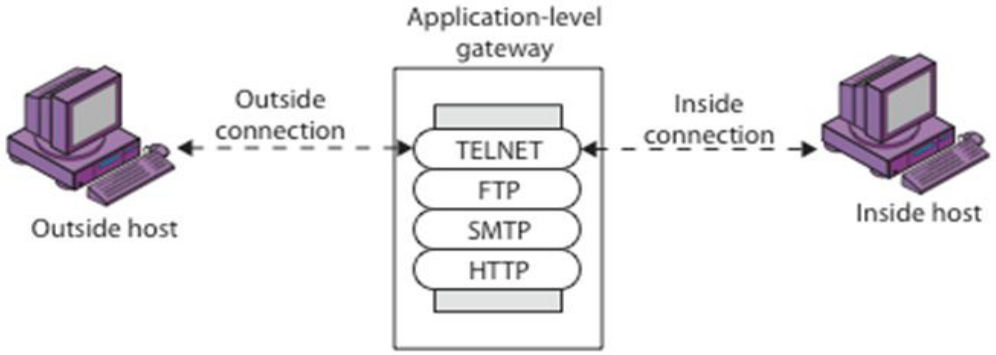
- ✓ The internal user of network contact to application gateway using HTTP or TELNET.
- ✓ The application gateway ask the user about remote host to which it wants to setup a connection.
- ✓ The application gateways ask for User ID and password to access the service on it.
- ✓ User provides User ID and password.
- ✓ The application gateway can access the remote host and pass the packets of users to remote host.
- ✓ Application gateway acts as a proxy of actual user.

Advantage:

- ✓ Application gateway is more secure than packet filter firewall because it simply detects a user is allowed to work or not.
- ✓ It can be used to log user activity and login.
- ✓ It has a significant impact on network performance.

Disadvantage:

- ✓ The disadvantage in terms of more time required to establish connectivity between application gateway and user.



4) Stateful multilayer inspection firewalls

- ✓ It combines the aspects of three types of firewall.
- ✓ They filter packets at Network Layer, determine whether session packets are legal and evaluate contents of packet at the application layer.
- ✓ It allows direct connection between client & host.
- ✓ They depend on algorithm to recognize and process application layer data instead of running application specific proxies.
- ✓ It offers high level of security, good performance, and transparency to end users.
- ✓ They are expensive and complex.

5) Next-generation Firewalls (NGFW)

- ✓ Many of the latest released firewalls are usually defined as 'next-generation firewalls'.
- ✓ This type of firewall is usually defined as a security device combining the features and functionalities of other firewalls.
- ✓ These firewalls include deep-packet inspection (DPI), surface-level packet inspection, and TCP handshake testing, etc.
- ✓ NGFW includes higher levels of security than packet-filtering and stateful inspection firewalls.
- ✓ Unlike traditional firewalls, NGFW monitors the entire transaction of data, including packet headers, packet contents, and sources.
- ✓ NGFWs are designed in such a way that they can prevent more sophisticated and evolving security threats such as malware attacks, external threats, and advance intrusion.
- ✓ NGFWs are a part of the process toward security virtualization because they can be software-based.
- ✓ Deep packet inspection and intrusion prevention systems are key parts of NGFWs that monitor and secure a network.
- ✓ NGFWs can monitor traffic at multiple levels of the OSI network model, but most importantly at Layer 7, the application layer.
- ✓ NGFWs can limit the amount of traffic they have to monitor by only allowing in traffic from white listed applications.

4.3 Intrusion detection systems (IDS):

Intruders: Intruder is anyone or anything that tries to get access to any part of your computer system. An intruder is typically referred to as a hacker.

Types of Intruders in Network Security

Computer intruders are basically divided into the following three categories:

Masquerader

These intruders are not authorized to use the system but still find ways to exploit the user's privacy and confidential information.

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Misfeasor

A misfeasor is authorized to use the system but misuses its access and privileges.

Clandestine User

These intruders have supervision powers over the entire system and misuse this control. Clandestine users typically use their power for financial gains.

Intrusion detection systems (IDS): It is a device or software application that monitor network or system activities for malicious activity and produce report on management system.

- ✓ It is a type of security management system for computers and networks.
- ✓ IDs gathers and analyzes information from various areas within a computer or a network to identify security threats.
- ✓ It includes functions like
    1. Monitoring and analyzing both user and system activities
    2. Analyzing system configurations and vulnerabilities
    3. Assessment of system and file integrity
    4. Ability to recognize patterns of attacks
    5. Analysis of abnormal activity patterns
    6. Tracking user policy violations

Components of IDS:

IDs consists of:

1. Event Generator
2. Analyzer
3. Response Module

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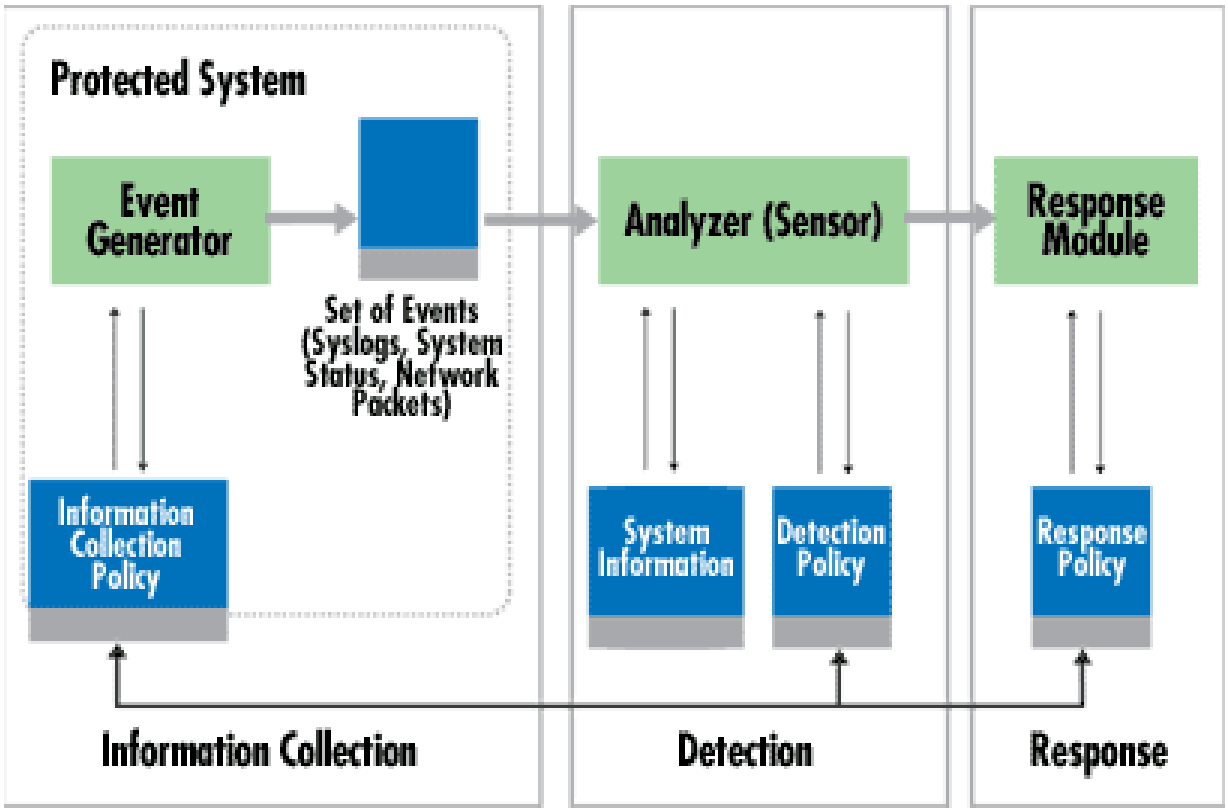


1) **Event Generator:**The event generator (operating system, network, application) produces a policy-consistent set of events that may be a log (or audit) of system events, or network packets.
2) **Analyzer:**The role of the sensor is to filter information and discard any irrelevant data obtained from the event set associated with the protected system.Sensor receive raw data from own IDs knowledge base,syslog,audit trails etc.
3) **Response Module:**Response module will fire the alarm if any intrusion of policy is detected by sensor.

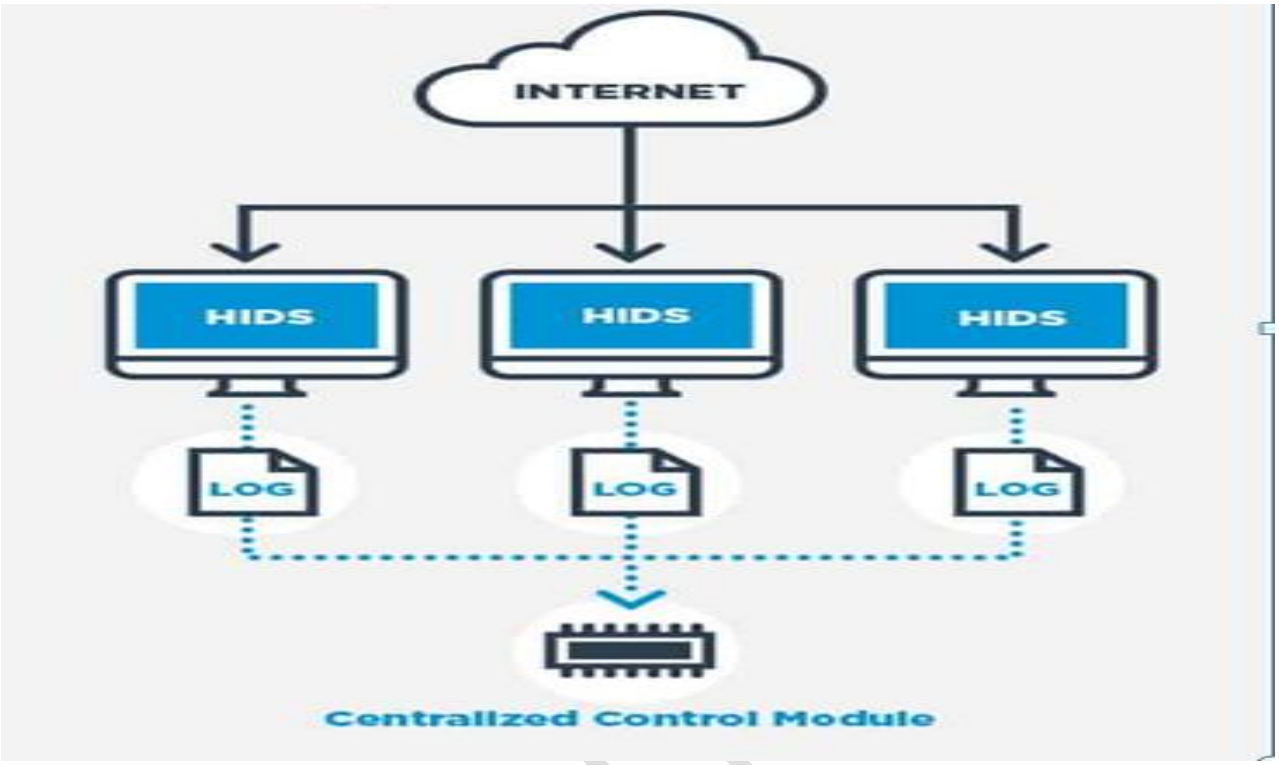
There are two types of Intrusion detection systems:

- ✓ Host based Intrusion detection systems
- ✓ Network based Intrusion detection systems

Host based IDS:

- ✓ HIDS run on individual hosts or devices on the network.
- ✓ It monitors and analyzes the internals of a computing system, network packets on its network.
- ✓ A HIDS monitors the inbound and outbound packets from the device only and will alert the user or administrator if suspicious activity is detected.
- ✓ It takes a snapshot of existing system files and matches it to the previous snapshot.
- ✓ If the critical system files were modified or deleted, an alert is sent to the administrator to investigate.
- ✓ An example of HIDS usage can be seen on mission critical machines, which are not expected to change their configurations.

- ✓ HIDS monitor state of a system, stored information, whether in RAM , in the file system, log files or else and check that the contents of these are expected(nit changed by intruders)



Advantages of HIDS

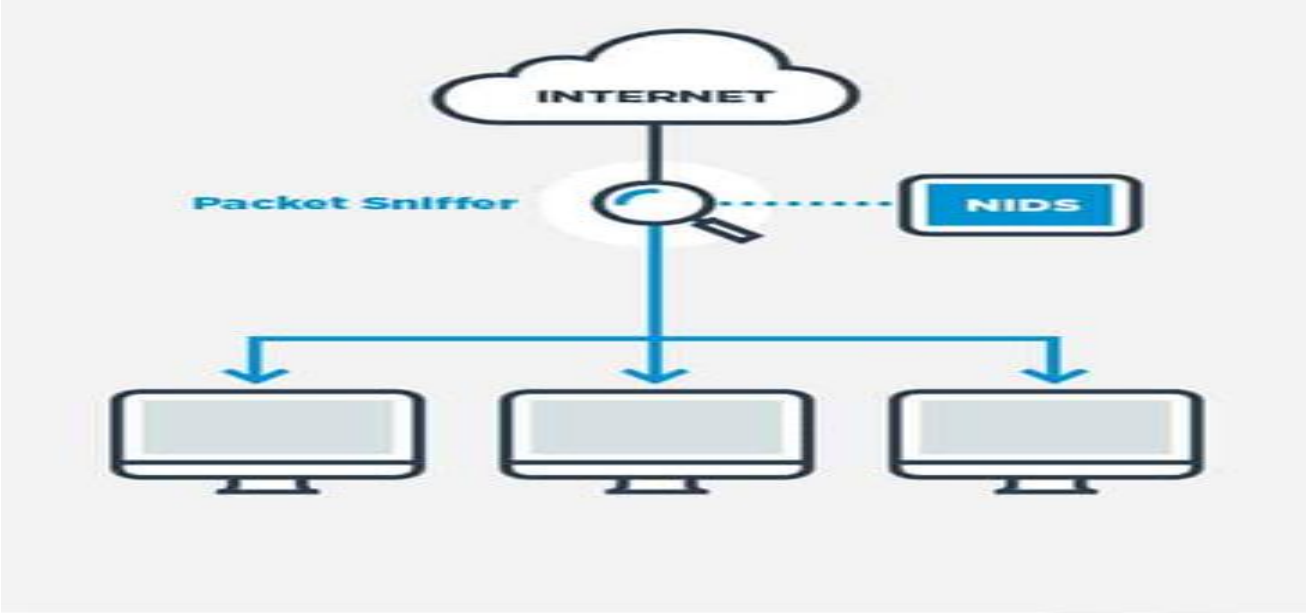
- ✓ Verifies success or failure of an attack: HIDS use system log which contains events that have actually occurred. They can determine whether an attack occurred or not.
- ✓ Monitors system activities: HIDS monitors user and file access activity including file accesses, changes to file permissions, attempts to install new executables etc.
- ✓ Detects attack that a NIDS fails to detect. Example: if an unauthorized user makes changes to system files from the system console, this kind of attack goes unnoticed by network sensors.
- ✓ Lower cost: HIDS sensors are cheaper than the NIDS sensors.

Disadvantages of HIDS

- ✓ HIDs are harder to manage, because information must be configured and managed for every host.
- ✓ The information sources for HIDS reside on the host. The IDS may be attacked and disabled as part of the attack.
- ✓ HIDs are not well suited for detecting network scans.
- ✓ HIDS can be disabled by certain denial of service attacks.

Network based IDS:

- ✓ NIDS are placed at a strategic point or points within the network to monitor traffic to and from all devices on the network.
- ✓ NIDS reads all inbound packets and searches for any suspicious patterns.
- ✓ It performs an analysis of passing traffic on the entire subnet, and matches the traffic that is passed on the subnets to the library of known attacks.
- ✓ Once an attack is identified, or abnormal behavior is sensed, the alert can be sent to the administrator.
- ✓ Example: NIDS is installing on the subnet where firewalls are located in order to see if someone is trying to break into the firewall.
- ✓ Ideally one would scan all inbound and outbound traffic, however doing so might create a bottleneck that would impair the overall speed of the network.
- ✓ OPNET and NetSim are commonly used tools for simulation network intrusion detection systems.



Advantages of NIDS

- ✓ A few well placed NIDs can monitor a large network.
- ✓ NIDs are usually a passive devices that listen on a network wire without interfering with the normal operation of a network.
- ✓ NIDs can be made very secure against attack and even made invisible to many attackers.

Disadvantages of NIDS

- ✓ NIDs may have difficulty to control all packets in a large or busy network and therefore, may fail to recognize an attack launched during period of high traffic.
- ✓ Many advantages of NIDs don't apply to more modern switch based network.
- ✓ NIDs cannot analyze encrypted information.
- ✓ Most NIDs can only find that an attack was initiated. But cannot tell whether or not an attack was successful.